Symmetric List Objects

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joint work with
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MSFP @ FLoC 2026, Lisbon, Portugal

July 18, 2026

Lists

In scheme:

```scheme
(a b c) ; a list of three elements
(a . (b . (c . ()))) ; the same list
() ; the empty list
(car '(a b c)) ; a
(cdr '(a b c)) ; (b c)
```

In Haskell:

```haskell
data [a] = [] | a : [a]
```

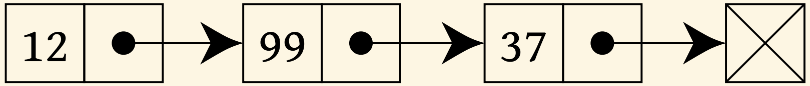
In Agda:

```agda
data 𝓛ist (A : Set) : Set where
  [] : 𝓛ist A
  _∷_ : A → 𝓛ist A → 𝓛ist A
```

In Rocq:

```coq
Inductive list (A : Type) : Type :=
  | nil : list A
  | cons : A → list A → list A.
```

Linked lists:



Constructions

In categories with finite products and countable coproducts, we can construct lists as:

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We can also construct lists as initial algebras of the following functor:

$$F(X) = 1 + A \times X$$

but this requires an initial object, and for the colimit of the following ω-chain to exist:

$$0 \to F(0) \to FF(0) \to FFF(0) \to \dots$$

List Objects

Lawvere introduced[1] natural numbers objects in his axiomatisation of set theory (ETCS):

$$
\begin{tikzcd}
 & N \arrow[dd, dashed, "\mathrm{rec}(z, s)"] & N \arrow[l, "\mathrm{succ}"'] \arrow[dd, dashed, "\mathrm{rec}(z, s)"] \\
1 \arrow[ur, "\mathrm{zero}"] \arrow[dr, "z"'] & & \\
 & X & X \arrow[l, "s"]
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[1] F. W. Lawvere, “An Elementary Theory of the Category of Sets*,” *Proceedings of the National Academy of Sciences*, vol. 52, no. 6, pp. 1506–1511, Dec. 1964, doi: 10.1073/pnas.52.6.1506.

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We can adapt this universal property to axiomatise list objects in general monoidal categories.

[1] F. W. Lawvere, "An Elementary Theory of the Category of Sets*," *Proceedings of the National Academy of Sciences*, vol. 52, no. 6, pp. 1506–1511, Dec. 1964, doi: 10.1073/pnas.52.6.1506.

List Objects

Definition 1 (List objects)

$$\begin{tikzcd}
& L(A) \arrow[dd, dashed, "\mathrm{it}(n, c)"] & A \otimes L(A) \arrow[l, "\mathrm{cons}"'] \arrow[dd, dashed, "\mathrm{it}(n, c)"] \\
1 \arrow[ur, "\mathrm{nil}"] \arrow[dr, "n"'] & & \\
& L & A \otimes L \arrow[l, "c"]
\end{tikzcd}$$

A list object on A is:

- an object $L(A)$ with maps $\mathrm{nil} : 1 \to L(A)$ and $\mathrm{cons} : A \otimes L(A)$, such that;
- for every L with maps $n : 1 \to L$ and $c : A \otimes L \to L$, there exists a unique *iterator*

$$\mathrm{it}(n, c) : L(A) \to L$$

such that the above diagram commutes.

Previous work

- Joyal introduced arithmetic universes with list objects
- Maietti[1] [2] studies them in type theory
- Paré and Roman[3] study monoidal categories with parametrized NNOs
- Cockett[4] and Jay[5] study pretoposes with list objects, called locoi
- Fiore and Saville[6] study parametrised list objects

[1] M. E. Maietti, "Joyal's Arithmetic Universes via Type Theory," *Electronic Notes in Theoretical Computer Science*, vol. 69, pp. 272–286, Feb. 2003, doi: 10.1016/S1571-0661(04)80569-3.

[2] M. E. Maietti, "Joyal's Arithmetic Universe as List-Arithmetic Pretopos," *Theory and Applications of Categories*, vol. 24, pp. 39–83, 2010, doi: 10.70930/tac/qee78enb.

[3] R. Paré and L. Román, "Monoidal Categories with Natural Numbers Object," *Studia Logica*, vol. 48, no. 3, pp. 361–376, Sep. 1989, doi: 10.1007/BF00370829.

[4] J. Cockett, "List-Arithmetic Distributive Categories: Locoi," *Journal of Pure and Applied Algebra*, vol. 66, no. 1, pp. 1–29, Oct. 1990, doi: 10.1016/0022-4049(90)90121-W.

[5] C. Jay, "Finite Objects in a Locos," *Journal of Pure and Applied Algebra*, vol. 116, no. 1–3, pp. 169–183, Mar. 1997, doi: 10.1016/S0022-4049(97)81630-1.

[6] M. Fiore and P. Saville, "List Objects with Algebraic Structure," *LIPIcs, Volume 84, FSCD 2017*, vol. 84, p. 16:1–16:18, 2017, doi: 10.4230/LIPICS.FSCD.2017.16.

Examples of list objects in nature

- In $\mathsf{Set}(1, \times)$ (or any Grothendieck topos over Set), list objects are cons-lists:

$$1 \xrightarrow{\text{nil}} \mathcal{L}_f(A) \xleftarrow{\text{cons}} A \times \mathcal{L}_f(A)$$

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- In $[\mathcal{C}, \mathcal{C}]_{\text{fin}}(\mathrm{Id}, \circ)$ (finitary endofunctors), list objects are the free monad construction:

$$\mathrm{Id} \xrightarrow{\text{nil}} F^*(F) \xleftarrow{\text{cons}} F \circ F^*(F)$$

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- In $\mathsf{Top}_\bullet(S^0, \wedge)$ (pointed spaces under smash product), list objects on connected spaces correspond to the James construction:

$$1 \xrightarrow{\text{nil}} J(A) \xleftarrow{\text{cons}} A \wedge J(A)$$

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2. These list objects also do not necessarily coincide with free monoids.
   - For example, every object of $\mathsf{Set}(0, +)$ is the free monoid on itself, but list objects on inhabited sets are always infinite:

$$\mu X.0 + (A + X) \quad \cong \quad \mu X.A + X$$

Parametrised List Objects

To solve the first problem, we strengthen the universal property as follows:

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1 \otimes P \arrow[r, "\mathrm{nil} \otimes P"] & L(A) \otimes P \arrow[d, dashed, "\mathrm{it}(n, c)"] & A \otimes L(A) \otimes P \arrow[l, "\mathrm{cons} \otimes P"'] \\
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String Diagrams

$$
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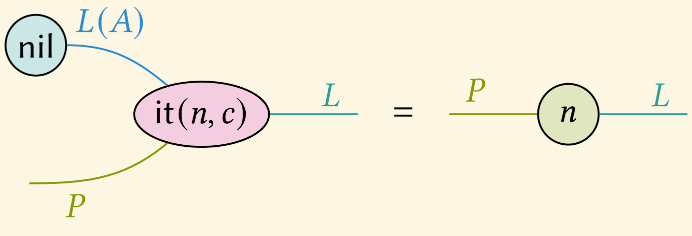
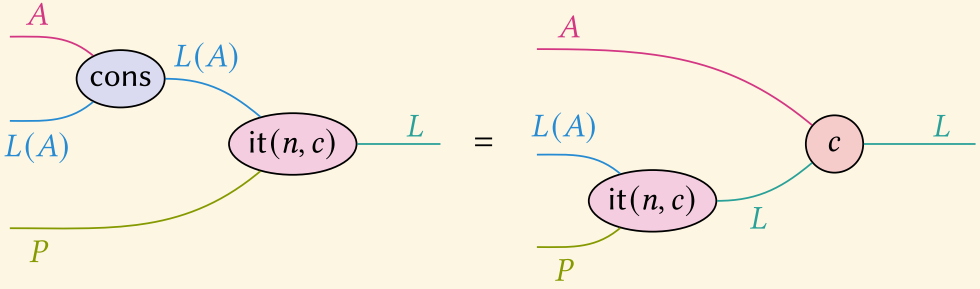


(a) nil-β



(b) cons-β

Using the iterator

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$$\mathrm{it}(n, c) : L(A) \otimes P \to L$$

- Since $+\!\!+ : L(A) \otimes L(A) \to L(A)$, we have $L(A)$ as both the parameter P and the target L.
- We now only need maps $n : L(A) \to L(A)$ and $c : A \otimes L(A) \to L(A)$.

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_++_ : List A → List A → List A
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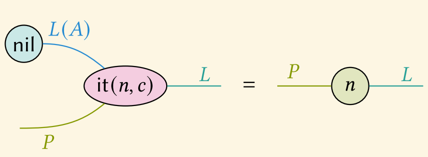

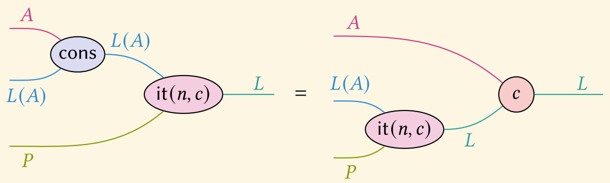

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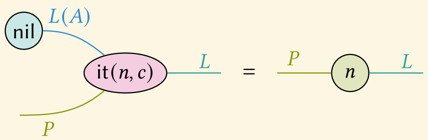

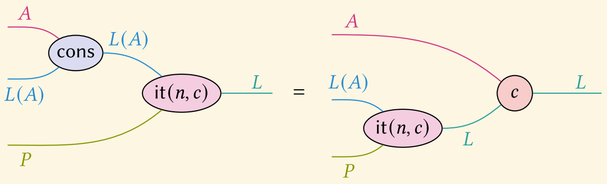

so we pick $n = \mathrm{id}$ and $c = \mathrm{cons}$:

$$+\!\!+ := \mathrm{it}(\mathrm{id}, \mathrm{cons}).$$

Monoids

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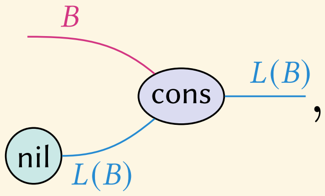
Lemma 2

If $\mathcal{C}$ admits parametrised list objects for every object, then $L(-)$ is a functor.

$$f : X \to Y \quad \mapsto \quad \left(X \xrightarrow{f} Y \xrightarrow{\eta_Y} L(Y)\right)^{\sharp} : L(X) \to L(Y)$$

$\eta :=$ 

, $\qquad \left(A \xrightarrow{g} (B, e, m)\right)^{\sharp} := \mathrm{it}(e, m \circ (g \otimes B))$

Monoids

$$\begin{tikzcd} \mathsf{Mon}(\mathcal{C}) \arrow[d, "U", "\dashv"'] \\ \mathcal{C} \arrow[u, bend left=60, "L"] \end{tikzcd}$$

Theorem 3 (M. Fiore and P. Saville[1])

If $(\mathcal{C}, \times, 1)$ is cartesian monoidal, then the induced monad is strong.

[1] M. Fiore and P. Saville, "List Objects with Algebraic Structure," *LIPIcs, Volume 84, FSCD 2017*, vol. 84, p. 16:1–16:18, 2017, doi: 10.4230/LIPICS.FSCD.2017.16.

Symmetric Lists

Hereafter, the ambient category is symmetric strict monoidal.

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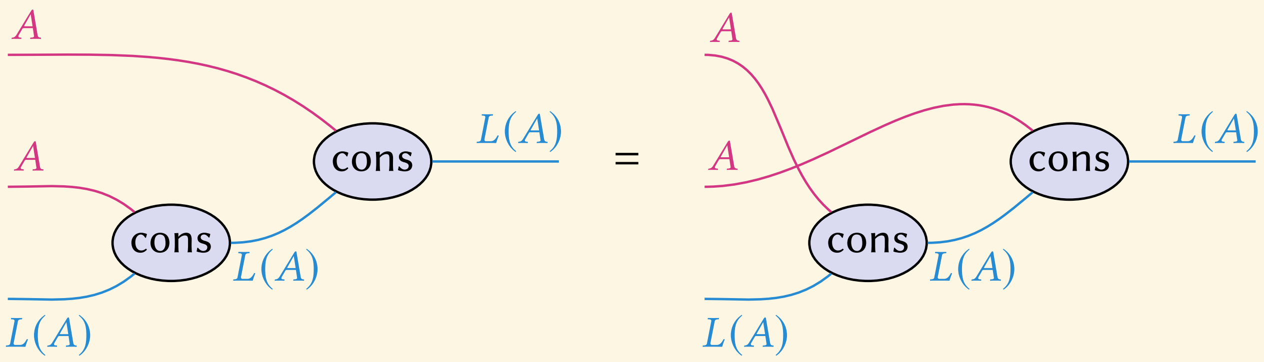

ilst axiom

A (*parametrised*) *symmetric list object* or *ilst* on A is a parametrised list object $L(A)$ satisfying the additional axiom above, and with the corresponding universal property.

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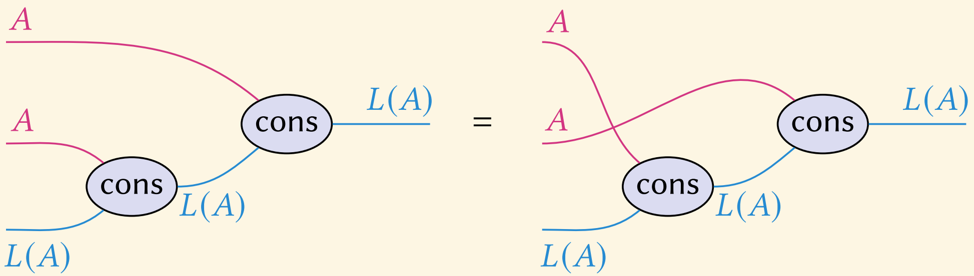


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data ℐlst (A : Set) : Set where
  [] : ℐlst A
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  swap : (a b : A) (xs : ℐlst A)
     → a :: b :: xs ≡ b :: a :: xs
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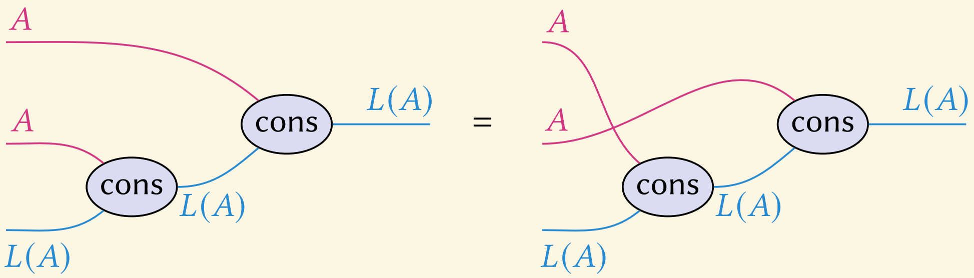


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$$\sum_{n \geq 0} A^n / S_n$$

Examples of symmetric list objects in nature

- In $\mathsf{Set}(1, \times)$ (or any Grothendieck topos over Set), symm. list objects are finite multisets:

$$1 \xrightarrow{\text{nil}} \mathcal{M}_f(A) \xleftarrow{\text{cons}} A \times \mathcal{M}_f(A)$$

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$$1 \xrightarrow{\text{nil}} \mathrm{SP}^\infty(X) \xleftarrow{\text{cons}} X \times \mathrm{SP}^\infty(X)$$

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- In $\mathsf{Rel}(1, \otimes)$ (binary relations with $\otimes$), symmetric list objects are finite multisets:

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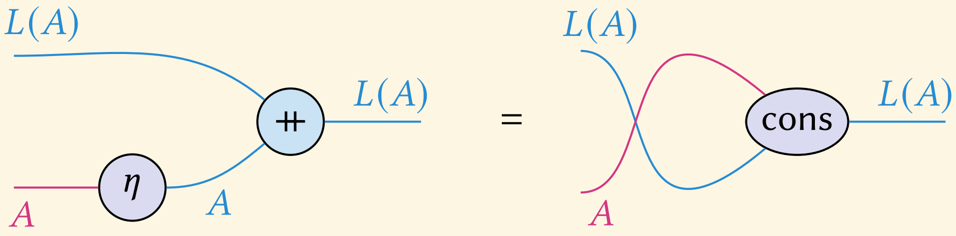
Symmetric Lists

Theorem 4

For any symmetric list $L(A)$:



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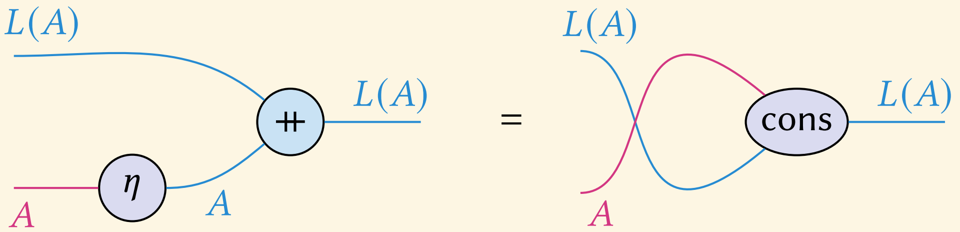

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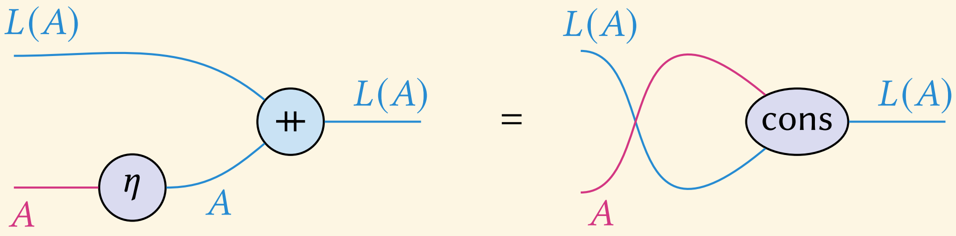

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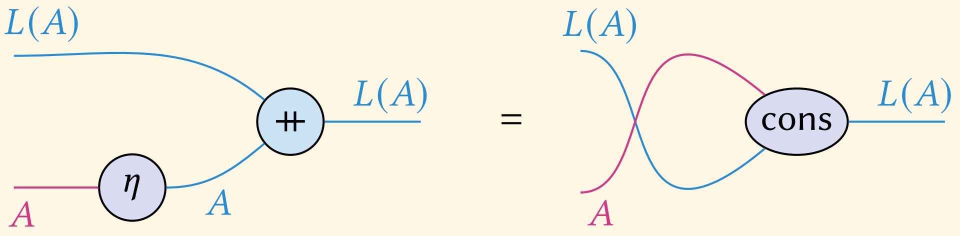

Proof. Let the two maps be f and g respectively.

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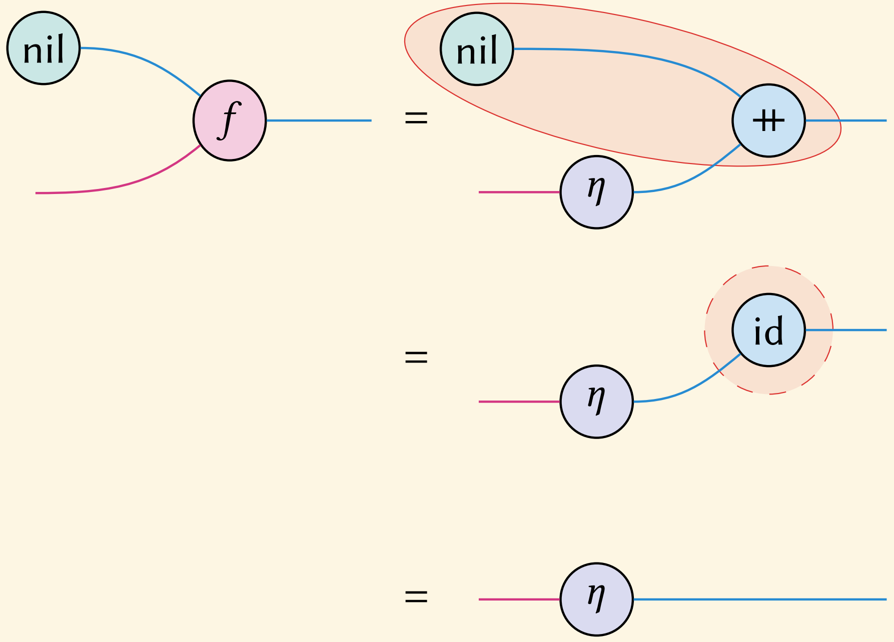
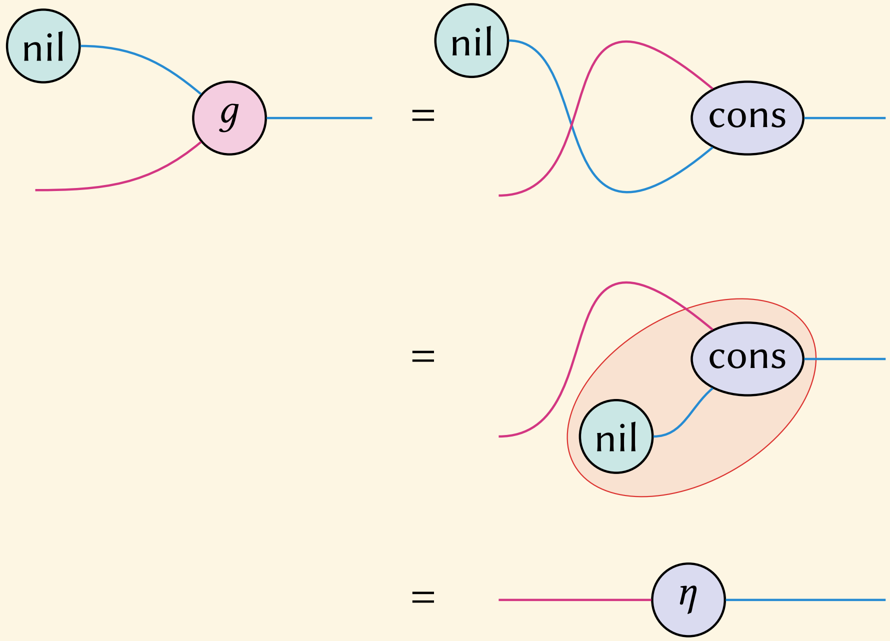
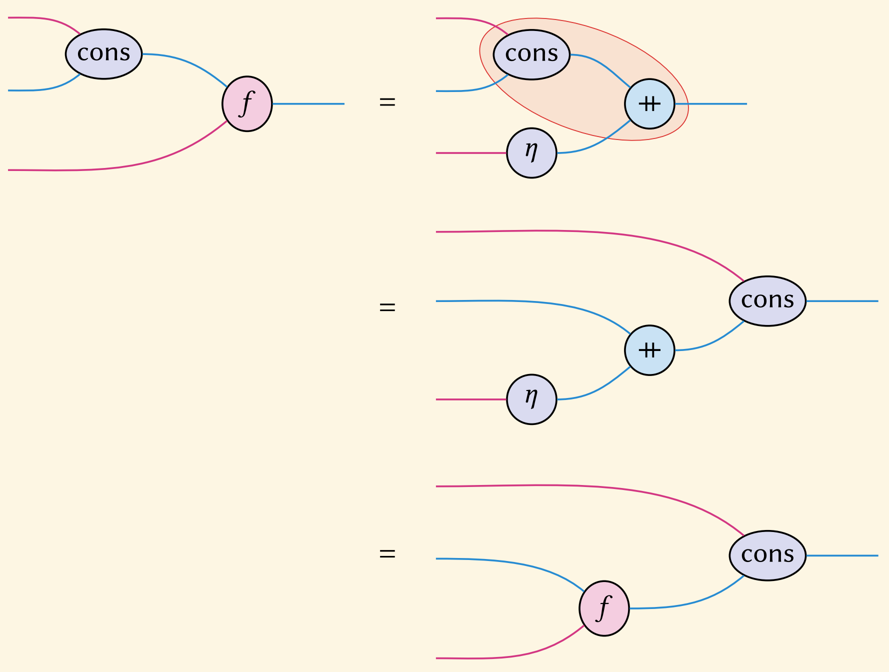
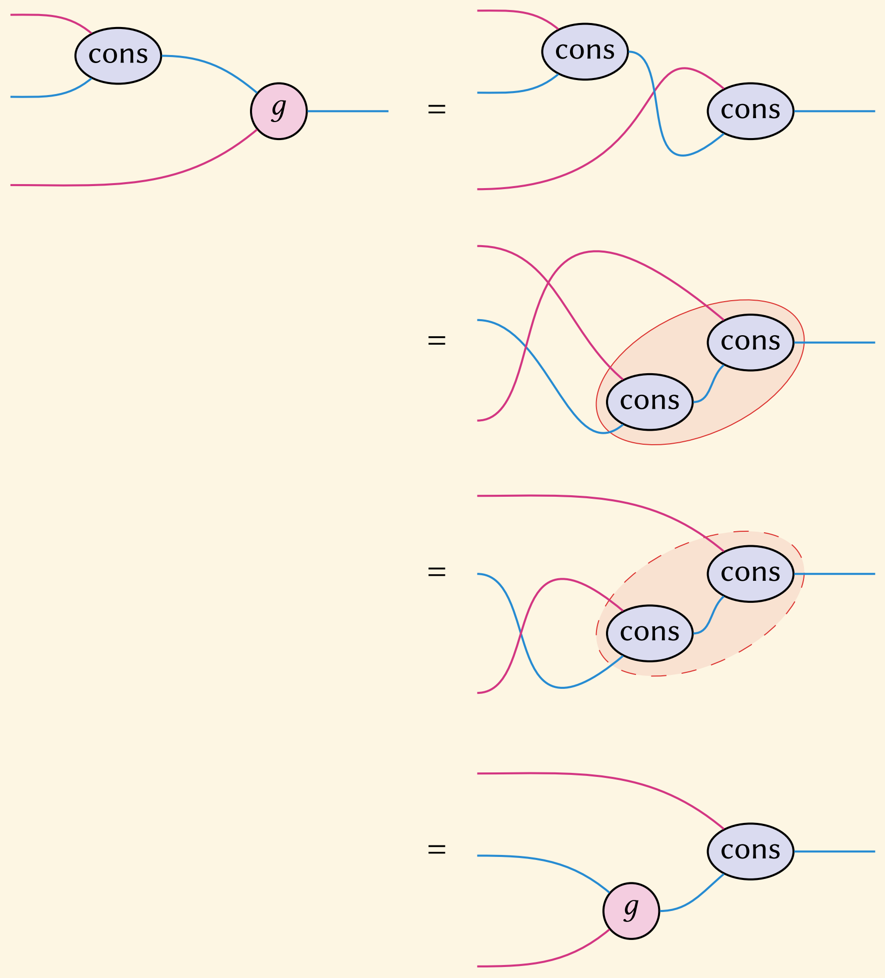
Since the iterator is unique, it suffices to show that they both satisfy the two computation rules nil-β and cons-β.


nil
f
=
nil
++
η
=
id
η
=
η


nil
g
=
nil
cons
=
cons
nil
=
η

□


cons
f
=
cons
++
η
=
cons
++
η
=
cons
f


cons
g
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cons
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=
cons
cons
=
cons
g

□

Symmetric Lists

Theorem 5

Let $(L(A), \mathrm{nil}, \mathrm{cons})$ be a list object on A. Then, the following are equivalent:

1. $(L(A), \mathrm{nil}, \mathrm{cons})$ is a symmetric list object;
2. $(L(A), \mathrm{nil}, +\!\!+)$ is a commutative monoid object;
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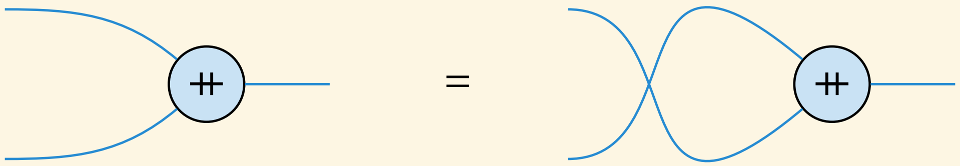
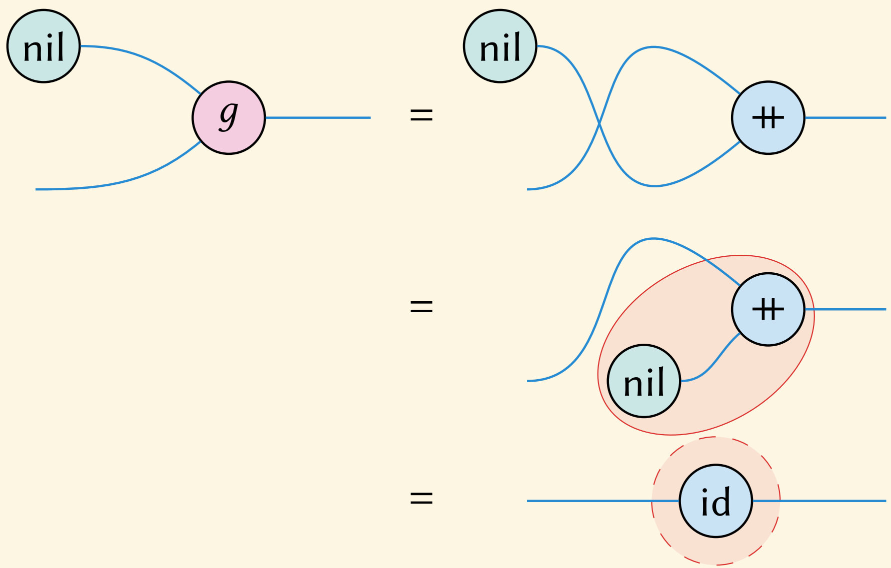
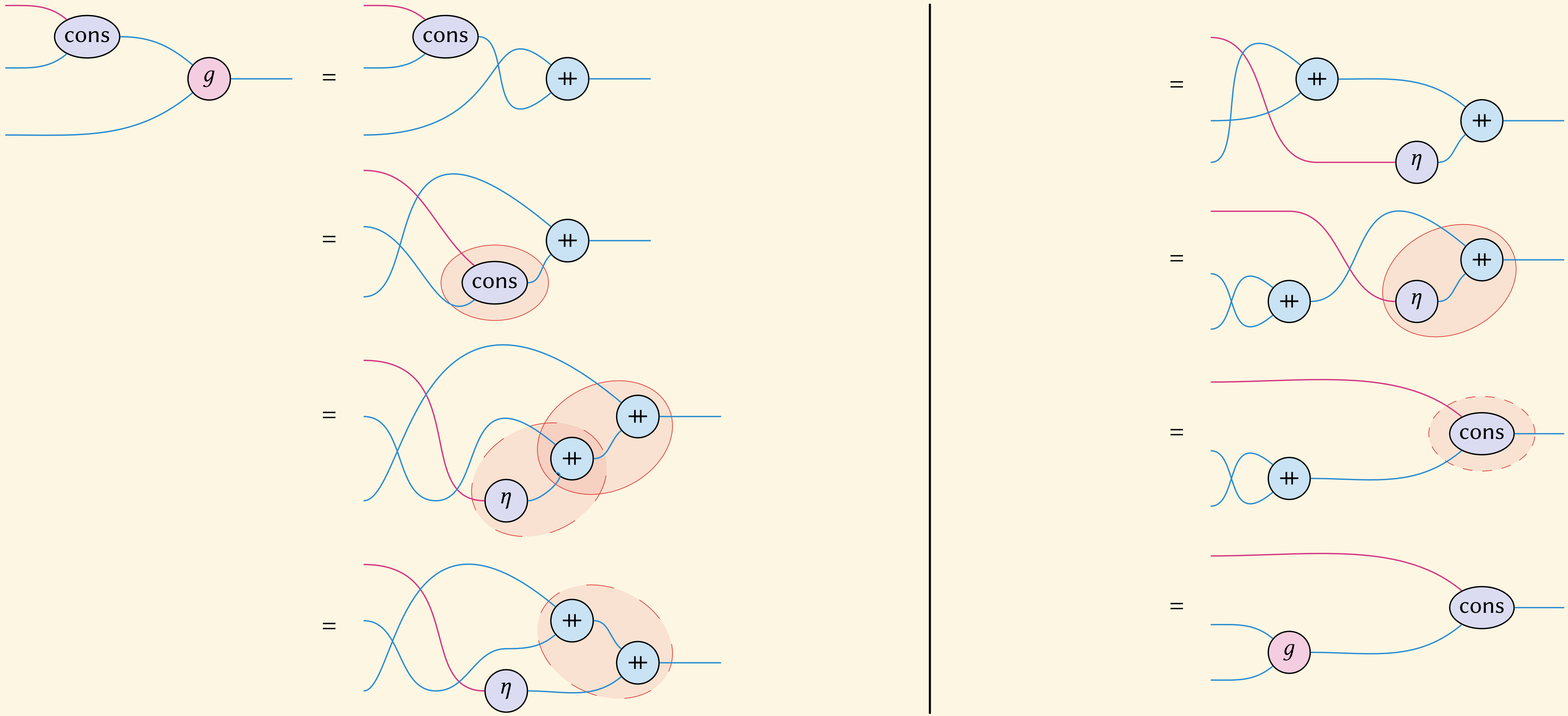
Proof. $(1 \Rightarrow 2)$ We already have that list objects are monoids, so we need only show that $+\!\!+$ is commutative:



Call the two maps f and g, respectively.

The left map f is $+\!\!+ := \mathrm{it}(\mathrm{id}, \mathrm{cons})$ by definition, so it suffices to show that the braided version satisfies the same computation rules:




cons
g
++
η

Lemma 5

If $\mathcal{C}$ admits symmetric list objects for every object, then $L(-)$ is a functor.

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$$\begin{tikzcd}
\mathsf{CMon}(\mathcal{C}) \arrow[d, "U"] \\
\mathcal{C} \arrow[u, bend left, "L"] \arrow[u, phantom, "\dashv"]
\end{tikzcd}$$

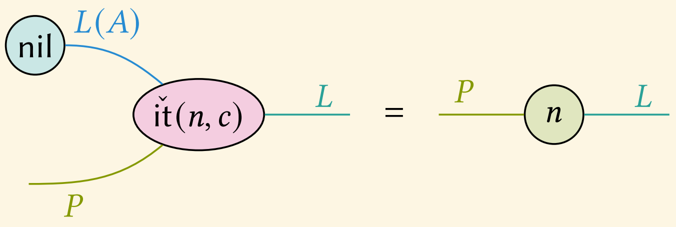
Theorem 6

If $(\mathcal{C}, \times, 1)$ is cartesian monoidal, and $L(A)$ is a list object, then $L(A)$ satisfies the following stronger universal property:

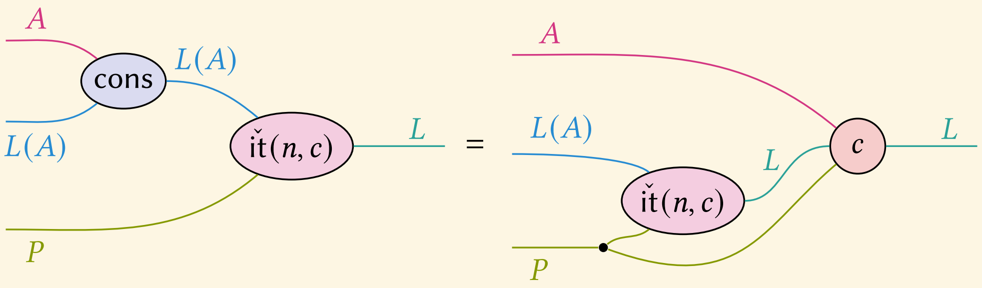
For every L with maps $P \xrightarrow{n} L \xleftarrow{c} A \times L \times P$, there exists a *doubly parametrised iterator*:

$$\check{\mathrm{it}}(n, c) : L(A) \times P \to L$$

such that the following computation rules hold:



(a) nil-β



(a) cons-β

Theorem 7

If $(\mathcal{C}, \times, 1)$ is cartesian monoidal, then the induced monad is strong, and further, commutative.

A construction of the right strength $\tau_{A,B} : L(A) \times B \to L(A \times B)$ is as follows.

We can equip $L(A \times B)$ with a doubly-parametrised list algebra structure using:

- $n := B \xrightarrow{!} 1 \xrightarrow{\mathrm{nil}} L(A \times B)$
- $c := A \times L(A \times B) \times B \xrightarrow{\cong} A \times B \times L(A \times B) \xrightarrow{\mathrm{cons}} L(A \times B)$.

From the strong universal property, we obtain a map

$$\tau_{A,B} := \check{\mathrm{it}}(n, c) : L(A) \times B \to L(A \times B)$$

$$
\begin{tikzcd}
& L(A) \times L(B) \arrow[dl, "\sigma_{L(A),B}"'] \arrow[dr, "\tau_{A,L(B)}"] & \\
L(L(A) \times B) \arrow[d, "L(\tau_{A,B})"'] & & L(A \times L(B)) \arrow[d, "L(\sigma_{A,B})"] \\
LL(A \times B) \arrow[dr, "\mu_{A \times B}"'] & & LL(A \times B) \arrow[dl, "\mu_{A \times B}"] \\
& L(A \times B) &
\end{tikzcd}
$$

$$
\begin{tikzcd}
 & & L(A) \times L(B) \arrow[dll, "\sigma_{L(A),B}"'] \arrow[drr, "\tau_{A,L(B)}"] & & \\
L(L(A) \times B) \arrow[d, "L(\tau_{A,B})"'] \arrow[ddrr, "\tau^{\sharp}_{A,B}"] & & & & L(A \times L(B)) \arrow[d, "L(\sigma_{A,B})"] \arrow[ddll, "\sigma^{\sharp}_{A,B}"'] \\
LL(A \times B) \arrow[drr, "\mu_{A \times B}"'] & & & & LL(A \times B) \arrow[dll, "\mu_{A \times B}"] \\
 & & L(A \times B) & &
\end{tikzcd}
$$

Conclusion

Summary

- Parametrised list objects give free monoids
- Symmetric list objects give free commutative monoids
- Iterator as recursion
- Doubly parametrised if Cartesian
- Strong monad if Cartesian
- Commutative monad if Symmetric

Future Work

- General theory of parametrised inital algebras
- Constructions of exponentials in LL
- Higher symmetric monoidal structures

Extended abstract with proofs